

IT5030 – Software Engineering Practices

Sprint 2 Report

6th March – 20th March 2026

Car Sales and Dealership Management System

Group 10

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Sprint Overview

Item	Details
Sprint Number	Sprint 2 – Customer Management, Sales & Appointments
Duration	6 th March – 20 th March 2026 (2 weeks)
Sprint Goal	Implement customer relationship management, sales transaction workflow with auto-invoicing, and appointment scheduling with conflict detection
Total Story Points Planned	21 points
Total Story Points Completed	21 points
Sprint Velocity	21 points / sprint

Table 1: Sprint 2 Summary

Sprint Backlog

The following user stories were selected for Sprint 2 during the Sprint Planning ceremony:

ID	User Story	Priority	Points	Status
US05	As a sales agent, I want to create and manage customer profiles so that I can track interactions and preferences	Must	5	Done
US06	As a sales agent, I want to record a vehicle sale with customer and payment details so that transactions are documented	Must	8	Done
US07	As a customer, I want to book a test drive appointment so that I can evaluate a vehicle before purchasing	Must	5	Done
US08	As a sales agent, I want to view my appointments and upcoming test drives so that I can prepare accordingly	Must	3	Done

Table 2: Sprint 2 Backlog Items and Status

Sprint Activities and Implementation

Customer Relationship Management (US05)

A dedicated customer management module was implemented to support the full lifecycle of customer interactions. The backend service provides CRUD operations with paginated search across name, email, and phone fields. The frontend renders a customer listing with search capabilities and detail views showing associated sales history and upcoming appointments. Input validation is enforced using Zod schemas on both client and server.

- Implemented POST/GET/PUT/DELETE `/api/customers` with pagination and search

- Built Customer List page with debounced search and pagination
- Built Customer Detail page showing linked sales and appointment history
- Built Customer Add/Edit form with react-hook-form and Zod validation
- Added email uniqueness validation and deletion safeguards (cannot delete if active sales exist)

Sales Transaction Workflow (US06)

The sales module delivers a complete transaction workflow driven by a finite state machine. Sales progress through PENDING, NEGOTIATION, and COMPLETED/CANCELLED states, with each transition triggering automatic side-effects on the vehicle inventory. Upon completion, an invoice is automatically generated with a 12% VAT calculation. All multi-table operations execute within Prisma transactions to guarantee data consistency.

- Implemented status machine: PENDING → NEGOTIATION → COMPLETED / CANCELLED
- Creating a sale automatically reserves the linked vehicle (status → RESERVED)
- Completing a sale marks the vehicle SOLD and generates an Invoice (amount + 12% tax)
- Cancelling a sale reverts the vehicle status to AVAILABLE
- Built Sales List with status-coloured badges and filtering capabilities
- Built Sale Detail page with status progression bar and contextual action buttons
- Built Create Sale form with searchable customer/vehicle dropdowns

Appointment Scheduling System (US07)

An appointment scheduling system was created supporting both test drive and consultation types. The backend implements double-booking prevention by checking for conflicting appointments within a one-hour window for both the vehicle and the assigned sales agent. All appointment date-times are validated to ensure they are in the future.

- Implemented POST/GET/PUT `/api/appointments` with conflict detection
- Prevented double-booking for same vehicle or same agent within 1-hour windows
- Enforced future-date-only constraint for new appointments
- Status transitions: SCHEDULED → COMPLETED / CANCELLED / NO_SHOW
- Built appointment list with type and status badge indicators

Role-Based Appointment Visibility (US08)

The appointment listing implements role-based data scoping: Sales Agents see only their own appointments by default, while Managers and Administrators can view all team appointments. Inline status update buttons on the appointment list allow agents to mark appointments as completed, cancelled, or no-show directly.

Dashboard Enhancements

The dashboard was enhanced with two additional summary cards: **Sales This Month** showing completed sales count, and **Upcoming Appointments** showing scheduled appointment count. The dashboard now displays six metrics providing a comprehensive operational overview.

Docker Deployment

The Docker configuration was updated with PostgreSQL health checks, automatic Prisma migration execution on container startup, and database seeding. The `docker-compose.yml` now ensures proper service startup ordering with the `depends_on: condition: service_healthy` directive.

Application Screenshots

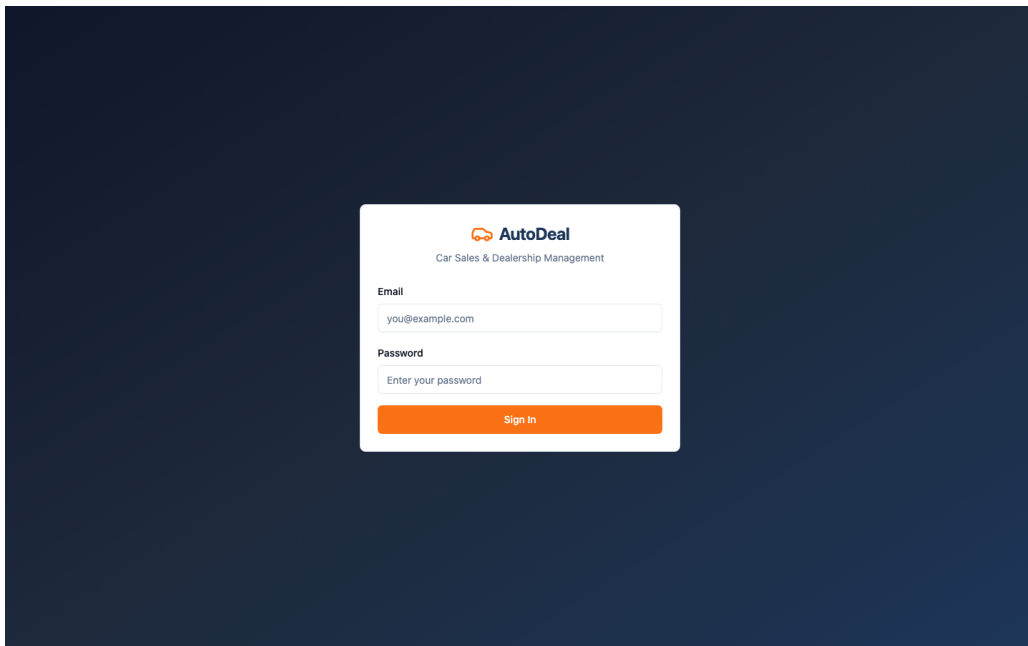


Figure 1: Login Page – Secure authentication with email and password

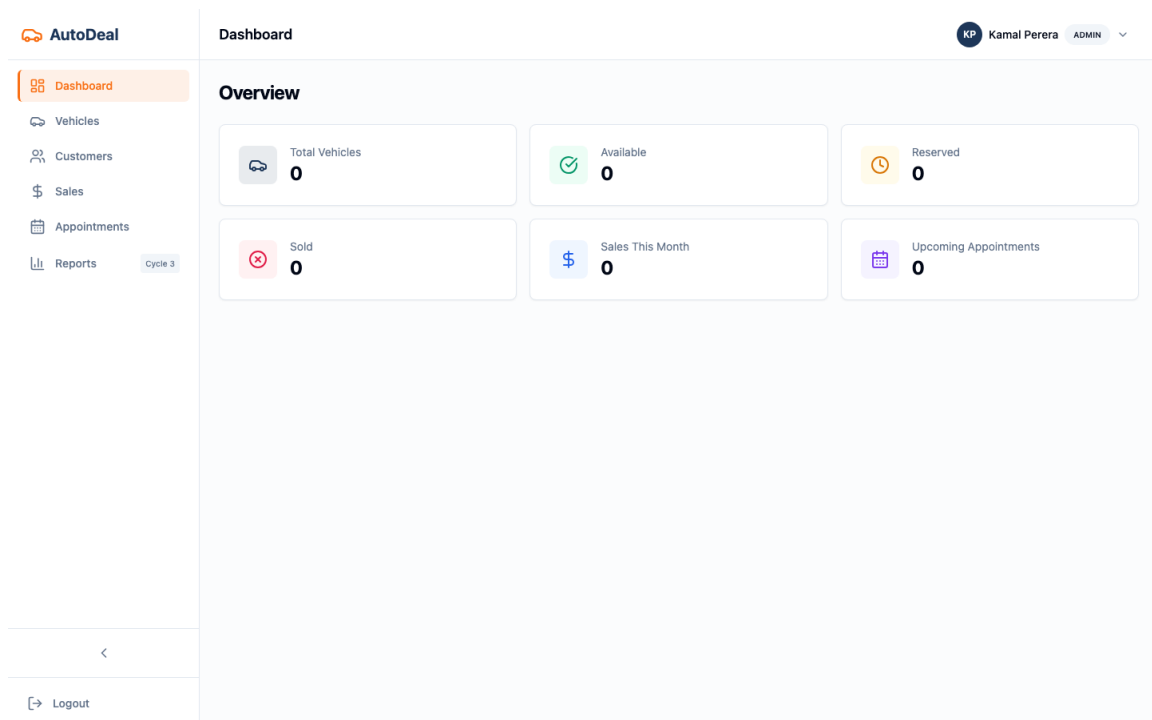


Figure 2: Dashboard – Overview with vehicle, sales, and appointment metrics

The screenshot displays the 'Vehicles' management interface in the AutoDeal system. The left sidebar contains navigation links for Dashboard, Vehicles (active), Customers, Sales, Appointments, and Reports. The top right shows the user 'Kamal Perera' with an 'ADMIN' role. The main content area features a search bar, filter buttons for Status, Fuel Type, and Transmissions, and a '+ Add Vehicle' button. A table lists 10 vehicles with columns for Make/Brand/Model, Year, VIN, Fuel Type, Price, Status, and Actions. The table includes pagination controls at the bottom.

MAKE / BRAND / MODEL	YEAR	VIN	FUEL TYPE	PRICE	STATUS	ACTIONS
Suzuki Alto K10	2020	J5AFNB135VA023456	PETROL	Rs 3800000	AVAILABLE	
Toyota Hilux Revo	2024	JTFHT02PXP0012345	DIESEL	Rs 25000000	AVAILABLE	
Honda Vezel Z	2023	HHGCV1F34NA023456	HYBRID	Rs 14200000	AVAILABLE	
Suzuki Swift RS	2023	J5AFNB135VA012345	PETROL	Rs 6500000	AVAILABLE	
Nissan X-Trail Ti	2024	JN8AT3MT9PW012345	PETROL	Rs 22000000	AVAILABLE	
Honda Civic RS	2022	HHGCV1F34NA012345	PETROL	Rs 9800000	RESERVED	
Nissan Leaf e+	2023	JN1BAZE12U0012345	ELECTRIC	Rs 11000000	AVAILABLE	
Toyota Aqua G	2021	JTDKN3DU5P0345678	HYBRID	Rs 7200000	SOLD	
Toyota Camry V6	2024	JTDKN3DU5P0234567	HYBRID	Rs 18500000	AVAILABLE	
Toyota Corolla GLi	2023	JTDKN3DU5P0123456	PETROL	Rs 12500000	AVAILABLE	

Figure 3: Vehicle Inventory – List view with search, filter, and pagination

The screenshot displays the 'Customers' management interface in the AutoDeal system. The left sidebar contains navigation links for Dashboard, Vehicles, Customers (active), Sales, Appointments, and Reports. The top right shows the user 'Kamal Perera' with an 'ADMIN' role. The main content area features a search bar and a '+ Add Customer' button. A table lists 5 customers with columns for Name, Email, Phone, Created, and Actions. The table includes pagination controls at the bottom.

NAME	EMAIL	PHONE	CREATED	ACTIONS
Dinesh Wijesinghe	dinesh.w@gmail.com	+94775678901	3/26/2026	
Sachini Bandara	sachini.b@gmail.com	+94776789012	3/26/2026	
Ruwan Wickramasinghe	ruwan.w@yahoo.com	+94777890123	3/26/2026	
Malini Senanayake	malini.s@outlook.com	+94778901234	3/26/2026	
Tharaka Gunawardena	tharaka.g@gmail.com	+94779012345	3/26/2026	

Figure 4: Customer Management – Customer list with search capabilities

AutoDeal

Dashboard

Vehicles

Customers

Sales

Appointments

Reports

Logout

Sales

KP Kamal PereraADMIN

+ New Sale

All Status

VEHICLE	CUSTOMER	AGENT	SALE PRICE	STATUS	DATE	ACTIONS
Nissan Ti	Malini Senanayake	Supun Silva	Rs 21,500,000	PENDING	3/22/2026	
Honda RS	Dinesh Wijesinghe	Supun Silva	Rs 9,500,000	NEGOTIATION	3/20/2026	
Toyota G	Ruwan Wickramasinghe	Supun Silva	Rs 7,000,000	COMPLETED	2/15/2026	

Figure 5: Sales Management – Sales list with status badges and filtering

Dashboard

Vehicles

Customers

Sales

Appointments

Reports

Logout

KP

Kamal Perera

ADMIN

Appointments

+ Schedule

All Status

All Types

DATE/TIME	TYPE	CUSTOMER	VEHICLE	AGENT	STATUS	ACTIONS
3/27/2026, 4:30:00 PM	CONSULTATION	Dinesh Wijesinghe	-	Supun Silva	SCHEDULED	
3/28/2026, 3:30:00 PM	TEST DRIVE	Sachini Bandara	Honda Z	Supun Silva	SCHEDULED	
3/29/2026, 7:30:00 PM	TEST DRIVE	Tharaka Gunawardena	Suzuki RS	Supun Silva	SCHEDULED	
3/30/2026, 2:30:00 PM	TEST DRIVE	Ruwan Wickramasinghe	Nissan e+	Supun Silva	SCHEDULED	

Figure 6: Appointment Scheduling – Appointment list with inline status updates

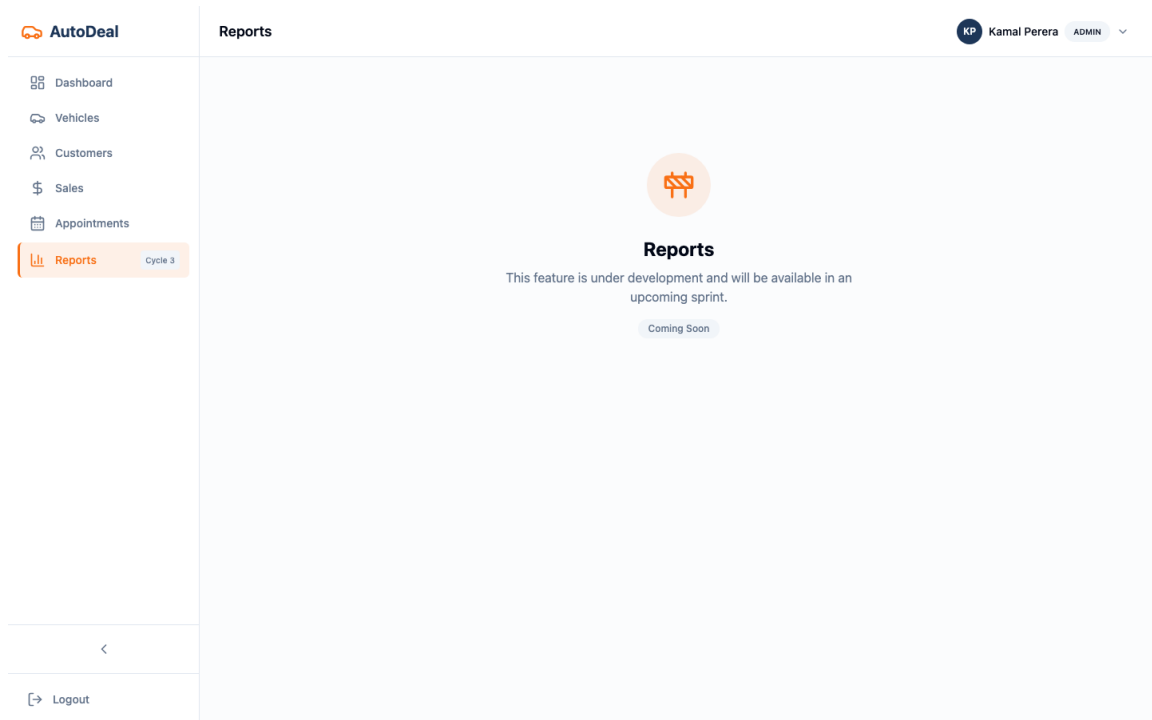


Figure 7: Coming Soon – Placeholder for features under development

Testing Activities and Results

Test Summary

Module	Test Type	Total	Passed	Failed
Authentication	Unit	8	8	0
Vehicle CRUD	Unit	8	8	0
Auth Middleware	Unit	6	6	0
Customer CRUD	Integration	5	5	0
Sales Workflow	Integration	6	6	0
Appointments	Integration	4	4	0
Total		37	37	0

Table 3: Sprint 2 Test Results Summary

Key Test Scenarios

- **Sales – Complete sale generates invoice:** Creating a sale and transitioning to COMPLETED automatically generates an invoice with correct tax calculation (12%). **Result: Passed.**
- **Sales – Cancel sale reverts vehicle:** Cancelling a NEGOTIATION sale correctly reverts the vehicle status from RESERVED to AVAILABLE. **Result: Passed.**
- **Appointments – Double-booking prevention:** Attempting to create overlapping appointments for the same vehicle returns a 409 conflict error. **Result: Passed.**
- **Customer – Delete with active sale:** Attempting to delete a customer with a PENDING sale returns a 400 error. **Result: Passed.**
- **Appointments – Role-based filtering:** SALES_AGENT sees only own appointments; ADMIN sees all. **Result: Passed.**

Scrum Artifacts

Task Board Summary

Category	To Do	In Progress	In Review	Done
Backend Tasks	0	0	0	11
Frontend Tasks	0	0	0	10
Testing/QA	0	0	0	4
Documentation	0	0	0	3
DevOps/CI	0	0	0	3
Total	0	0	0	31

Table 4: Sprint 2 Task Board at Sprint End

Burndown

All 21 planned story points were completed within the sprint. The team achieved a velocity of 21 points, an improvement over Sprint 1's 15 points, reflecting the established development patterns and improved team coordination.

Individual Contributions

Member	Role	Key Contributions	Effort
P.M.A.D.N. Kulathunga	Product Owner	Sprint 2 PRD authoring, acceptance criteria for US05–US08, sales status machine definition, RBAC matrix, Jira backlog management, sprint planning	22%
B.V.S.M. Athurupana	Scrum Master	Daily standup facilitation, Docker deployment configuration with health checks, GitHub Actions CI pipeline, sprint retrospective documentation	18%
R.R.M.G.H. Rathnayake	Frontend Dev	Customer list/detail/form pages, sales list/detail/create pages, appointment list/create pages, dashboard enhancements, Coming Soon page	22%
H.M.S.N. Rajapaksha	Backend Dev	Customer/Sales/Appointment/Invoice REST APIs, Prisma transaction-based sales workflow, double-booking prevention, auto-invoice generation	20%
G.G.P.S. Gunathilaka	Full-Stack/QA	Integration testing for sales workflow and appointments, smoke testing customer CRUD, code review, appointment conflict test coverage	18%

Table 5: Sprint 2 Individual Contributions

Sprint Review

Completed Features

- **Customer Management:** Full CRUD with search, pagination, and linked sales/appointment history display
- **Sales Workflow:** Complete transaction pipeline with status machine, automatic vehicle status synchronisation, and invoice auto-generation with 12% VAT
- **Appointment Scheduling:** Test drive and consultation booking with double-booking prevention for both vehicles and agents
- **Invoice System:** Automatic invoice creation on sale completion with amount, tax, and total calculations
- **Dashboard Enhancement:** Six summary cards showing vehicle counts, sales this month, and upcoming appointments
- **Docker Deployment:** Health checks, auto-migration, and seed data on container startup
- **CI Pipeline:** GitHub Actions with TypeScript checking, linting, and test execution

Incomplete / Carried Forward

- No items carried forward. All Sprint 2 user stories were completed successfully.

Sprint Retrospective

What Went Well

- Established development patterns from Sprint 1 significantly accelerated Sprint 2 development. The service-route-validator module structure enabled rapid feature addition.
- Prisma transactions ensured data consistency across the complex sales workflow involving vehicle status updates and invoice generation.
- Docker containerisation allowed all team members to develop against identical environments, eliminating “works on my machine” issues.

What Could Be Improved

- Frontend test coverage remains lower than backend. Sprint 3 should prioritise React Testing Library tests for critical user flows.
- The appointment scheduling UI would benefit from a calendar view rather than a simple list to better visualise scheduling conflicts.

Action Items for Sprint 3

#	Action Item	Owner
1	Implement analytics dashboard with recharts for sales performance and inventory trends	Frontend Dev
2	Build invoice PDF generation and inventory reporting endpoints	Backend Dev
3	Increase frontend test coverage to $\geq 60\%$ with React Testing Library	Full-Stack/QA

Table 6: Retrospective Action Items

Sprint 3 Preview

Sprint 3 Goal

Implement analytics dashboard with visual charts, sales agent performance tracking, invoice generation for completed sales, and inventory reporting.

Planned User Stories

ID	User Story	Priority	Points
US09	As a manager, I want to view a sales analytics dashboard so that I can monitor dealership performance	Should	8
US10	As a manager, I want to track sales agent performance and commissions so that I can manage the sales team effectively	Should	5
US11	As a sales agent, I want to generate invoices for completed sales so that customers receive proper documentation	Should	5
US12	As a manager, I want to view inventory reports showing stock levels and turnover rates	Should	5

Table 7: Sprint 3 Planned Backlog